LPS
APP
NADPH oxidase complex
JNK
NOS2
FOS
JUNB
AP-1 transcription factor
SRR
AP-1 : SRR
nucleus
nitric oxide
SNP
SNAP
dexamethasone
SRR steady state dimer
inactive SDS/βME resistant SRR dimer
peroxynitrite
superoxide
Spermine NONOate
HAPI microglia
NMDA glutamate receptor activity
neuron
IL1B
APOE
activated microglia M1
NOX1
CYBB
NCF2
NCF1
ARG1
CHI3L1
SLC7A11
PKA
dibutyryl cAMP
microglia
L-glutamate
AP5
NMDA receptor glutamate site
NMDA receptor glycine/D-serine site
DCKA
L-NIO
NOS3
20-HETE
vasodilation
D-serine
NP-EGTA
Ca2+
ACPD
Metabotropic Glutamate Receptor
prostaglandin E2
astrocyte activation
GLUD1
GLUD2
glutamate dehydrogenase
glutamate catabolic process
hydrogen peroxide
3-hydroxypyruvate
NH4+
D-glyceraldehyde 3-phosphate
NAD(+)
NADH
MPP+
GAPDH
GAPDH : SRR
ATP
glycolysis
ER stress response
neuron death
nucleocytoplasmic shuttling
DAO
H2O
O2
D-serine biosynthetic process
L-serine
glycine
glycine/D-serine site
motor activity
NMDA receptor
voltage-gated sodium channel activity
MK-801
synaptogenesis
TGFB1
6-OHDA
BME
etoposide
LMNA
SRR : LMNA : GAPDH
FBXO22
KDM4A
isoform 1
isoform 3
SRR : FBXO22a
SRR : FBXO22b
PRKCA
nucleus
motor cortex
APP
IL4
NCF1
MAPT
axonal varicosity
protofibril
axon
SYP
synapse
post synapse
cAMP
cyclic nucleotide phosphodiesterase
IBMX
caffeine
PHGDH
PHGDH tetramer
PSAT1
pyridoxal 5'-phosphate
PSAT1 : PXLP dimer
3-phospho-D-glyceric acid
L-glutamate(1−)
2-oxoglutarate(2−)
3-phosphonooxypyruvic acid
O-phospho-L-serine
PSPH
Mg2+
PSPH : Mg2+ dimer
phosphate(3−)
L-serine biosynthetic process
SRR : pyridoxal 5'-phosphate
astrocyte
vesicle
SLC1A4
axon guidance
striatal neuron
striatum
mitochondrial matrix
pyruvate dehydrogenase complex
BAX
CASP3
CYCS
staurosporine
DA neuron
substantia nigra
astrocyte activation
corpus callosum
KCl
late phase of apoptosis
dizocilpine
L-serine
met□phen
MAPK8
MAPK9
MAPK10
JNKs
apoptosis
UPS
granule neuron
cerebellum
brain
pyramidal neuron
CA fields
multiform layer
pyruvic acid
NH3
PICK1
SRR : PICK1
PICK1 : EPHA4
PICK1 : EPHB3
EPHB3
EPHA4
EFNB3
glutamine
SNARE
vesicle fusion
astrocytic vesicle
D-isoleucine
SLC7A10
antiporter activity
long-term synaptic potentiation
induction of synaptic plasticity by chemical substance
CA1
hippocampus

L-serine
SRR
pyramidal neuron
D-serine
SLC17A7
SLC17A6
TGFB1
caffeine
BME
cyclic nucleotide phosphodiesterase
voltage-gated sodium channel activity
glycine
D-serine
glycine/D-serine ratio
mortor activity
microglia
IL1B
dibutyryl cAMP
PKA
APOE
NCF2
CHI3L1
SLC7A11
ARG1
NOX1
CYBB
NCF1
activated microglia M1
cAMP
MAPT
cortex
L-glutamate
protofibril
SYP
PRKCA
FBXO22
isoform 1
isoform 2
SRR : FBXO22b
SRR : FBXO22a
nucleocytoplasmic shuttling
ER stress response
neuron death
etoposide
MK-801
NMDA receptor
synaptogenesis
FOS
GAPDH
LMNA
KDM4A
nucleus
SRR : LMNA : GAPDH
D-serine biosynthetic process
post synapse
synapse
DCKA
AP5
NMDA receptor glycine/D-serine site
NMDA receptor glutamate site
NMDA glutamate receptor activity
L-NIO
arteriole
NOS3
20-HETE
vasodilation
ACPD
NP-EGTA
TGFB receptor signaling pathway
Metabotropic Glutamate Receptor
Ca2+
astrocyte activation
V-ATPase
VAMP2
H+
proton transmembrane transport
vesicle
synaptic vesicle lumen acidification
Cl-
prostaglandin E2
glutamate catabolic process
GLUD1
GLUD2
glutamate dehydrogenase
D-glyceraldehyde 3-phosphate
NAD(+)
3-phospho-D-glyceroyl dihydrogen phosphate
glycolysis
SLC1A4
SRR : ATP
ATP
GAPDH : SRR
MPP+
NADH
PHGDH
L-serine biosynthetic process
O2
DAO
hydrogen peroxide
3-hydroxypyruvate
NH4+
staurosporine
BCL2
BAX
apoptosis
CASP3
CYCS
Mn2+
pyridoxal 5'-phosphate
SDS
6-OHDA
serine racemase activity
SRR : pyridoxal 5'-phosphate
negative regulation of oxidative stress-induced neuron death
pyruvate dehydrogenase complex
acetyl CoA
3-phospho-D-glyceric acid
PHGDH tetramer
PSAT1
3-phosphonooxypyruvic acid
L-glutamate(1−)
astrocyte
O-phospho-L-serine
PSPH
Mg2+
PSPH : Mg2+ dimer
2-oxoglutarate(2−)
phosphate(3−)
neuron development
microcephaly
axon guidance
SLC1A3
13J
CA1
striatal neuron
neuron
SRR : pyridoxal 5'-phosphate
APP
JNK
AP-1 transcription factor
AP-1 : SRR
dexamethasone
SNP
NADPH oxidase complex
SNAP
NOS2
nitric oxide
superoxide
peroxynitrite
inactive SDS/βME−resistant SRR dimer
SRR steady−state dimer
dizocilpine
met−phen
KCl
granule neuron
LPS
H2O
pyruvic acid
NH3
MAPK8
MAPK9
MAPK10
JNKs
late phase of apoptosis
Spermine NONOate
SNARE
vesicle fusion
EFNB3
EPHA4
PICK1
EPHB3
PICK1 : EPHA4
PICK1 : EPHB3
SRR : PICK1
glutamine
SLC7A10
antiporter activity
induction of synaptic plasticity by chemical substance
D-isoleucine
long-term synaptic potentiation
NMDA glutamate receptor activity